steadily increasing to meet the faster response times, higher precision, and greater overall performance required by many emerging applications. The RISC-V architecture makes it easy to add cores to the driver ICs that are responsible for the communication and protocols, offering more complete solutions and higher levels of integration.

The road ahead

As the RISC-V and other early open source silicon platforms emerge, their development ecosystems will get a head start from the rich trove of resources already created by the open software and hardware movements. By enabling small, agile groups of entrepreneurs to bring cutting-edge products to market quickly and in a cost-effective manner, these open source technologies will continue to democratise and accelerate innovation within the tech economy. END

References

[1] Lee Goldberg, 'Maker Technology's Stone Soup Alchemy' – Texas Instrument's E2E Blog; *https://e2e.ti.com/blogs_/archives/b/launchyourdesign/archive/2014/11/11/maker-technology-39-s-stone-soup-alchemy*
[2] Carl Shapiro and Hal R. Varian (1999), Information Rules. Harvard Business School Press; ISBN 0-87584-863-X (courtesy of Wikipedia)
[3] Modern open source software probably owes its existence to Richard Stallman, a researcher at the MIT Artificial Intelligence Laboratory, and creator of an operating system named GNU, a recursive acronym for 'GNU's Not Unix!' an open, free alternative to AT&T's UNIX. Its first general release was in 1989.
[4] Wikipedia entry on GNU; *https://en.wikipedia.org/wiki/GNU*
[5] Dan Price, 'The True Market Shares of Windows vs Linux Compared' - MakeUseOf , March 27, 2018; *https://www.makeuseof.com/tag/linux-market-share/*
[6] *https://www.arduino.cc/en/Guide/Introduction*
[7] Lee Goldberg, 'Arduino Open Source Platform Unleashes Creativity' Nov. 23, 2011, Digi-Key TechZone; *https://www.digikey.com/en/articles/techzone/2011/nov/arduino-open-source-platform-unleashes-creativity*
[8] RISC-V Foundation; *https://riscv.org/*

By: Lee Goldberg and Jonas Proeger

Lee Goldberg is a Si-Tech journalist, a self-identified "recovering engineer". He is currently working as a freelance writer and a contributor to several publications. He can be reached at *lgoldberg@green-electronics.com*.

Jonas P. Proeger has been running the marketing team at TRINAMIC Motion Control since 2012. He deeply believes in the concept of open source software and hardware. He successfully launched the open source 3D printer cape 'T-Bone' on Indiegogo in 2013.

Continued from page no. 35...

Sahana is FOSS that aims to provide comprehensive solutions to information management in relief operations, recovery and rehabilitation. It ensures:

- An open access approach
- Low-cost deployment
- Easy adaptability

Sahana was developed by a group of volunteers in Sri Lanka during the Indian Ocean tsunami.

InaSAFE is used in Asia for disaster management. It was developed jointly by developers in Indonesia, Australia and the World Bank. It produces realistic natural hazard impact scenarios for better planning, preparedness and response activities. It provides a simple but rigorous way to combine data from scientists, local governments and communities to provide insights into the likely impacts of future disasters.

Even Facebook is assisting disaster victims to a great extent. Its disaster map provides information about where populations are located, how they are moving and where they are checking in for safety, during a natural disaster.

Disaster management and Big Data

Big Data offers new options for natural disaster management, and can help to visualise, analyse and predict natural disasters. For example, satellite remote sensing technology helps to assess the damage caused by a disaster, can forecast storms, and so on.

Big Data can help to save humans from all kinds of disasters. It can coordinate relief efforts for those who are the most vulnerable. Both crowd-sourcing of data and social media data are contributing greatly in disaster management. Government agencies, though, have been slow in launching their own crowd-sourcing platforms for disaster management.

Social media platforms, including Twitter, YouTube, Foursquare and Flickr, have been contributing significantly in disaster management. Geo-tagged social media data can be collected by streaming the harvest from the APIs provided by the social media companies. Mobile GPS has emerged as an effective means of gathering mobile sensing data because it can be used to detect human mobility and behaviour during large scale natural disasters. Predictive analytics can be a powerful tool for natural disaster management. Sensor data is usually the major source for developing an early warning system. END

References

[1] *www.researchgate.net*
[2] An article on disaster management by Sachi Pandey and others
[3] 'Disaster risk management in Asia and Pacific' by Ian Davis
[4] *https:// timreview.ca/article*

By: Vinayak Ramachandra Adkoli

The author is a B.E. in industrial production. He has worked as a lecturer in the mechanical department for ten years in three different polytechnics. He is also a freelance writer and cartoonist. He can be contacted at *karnatakastory@gmail.com*.